Exercise sheet 5

Exercise 1 *Give an example of matrix M with integer coefficients of determinant 1 such that M has no root of unit as eigenvalues, but M is not hyperbolic*

Exercise 2 *Let T be a hyperbolic linear bijection on $\mathbb{R}^n$ and $\|\cdot\|$ a norm on $\mathbb{R}^N$ adapted to T and $\epsilon > 0$. Show that for all sequence $(y_n)_{n\in\mathbb{Z}}$ in $\mathbb{R}^N$ with*

$$\sup_{n\in\mathbb{Z}} \|y_{n+1} - T(y_n)\| \leq \epsilon,$$

then there exists a unique point $y \in \mathbb{R}^N$ such that

$$\sup_{n\in\mathbb{Z}} \|y_n - T^n(y)\| \leq \frac{\epsilon}{1 - ch(T)}$$

(Pseudo-orbit of T is close to a real orbit)